The Linear Algebra of Well-Posedness

For a linear transformation $A: V \to W$,

(*) solve $Ax = b$ for $x \in V$

given "forcing" $b \in W$.

"Well-posed"

1) Solution exists
2) Solution is unique
3) Solution depends continuously on data (on A and b)

"Four Fundamental Sub."

$N(A) = \{x \in V : Ax = 0\}$

$R(A) = \{Ax \in W : x \in V\}$

$N(A^*) = \{x \in W : A^*x = 0\}$

$R(A^*) = \{A^*x \in V : x \in W\}$

Adjoint $A^*: W \to V$ s.t.

$\langle Ax, y \rangle = \langle x, A^*y \rangle$

"Generalizes matrix transpose"

Today: Study connection between 4 F.S.'s and criteria for well-posedness using Fourier Series.

Diff. Ops. with Constant Coeffs

Consider a general const. coeff. differential operator:

$$A = a_0 I + a_1 \frac{d}{dx} + \cdots + a_m \frac{d^m}{dx^m}$$

acting on smooth periodic function on $[0,1]$.

In the Fourier basis, $e_k(x) = e^{2\pi i k x}$ for $k = 0, \pm 1, \pm 2, \ldots$

$$[A e_k](x) = [a_0 + a_1 \underbrace{(2\pi i k)^0}_{1} + a_1 (2\pi i k)^1 + \cdots + a_m (2\pi i k)^m] e_k(x)$$

"Characteristic Polynomial"

$$= p_A(2\pi i k)$$

$$p_A(\lambda) = a_0 + a_1 \lambda + \cdots a_m \lambda^m$$

The Fourier modes are eigenvectors of A,

$$A e_k = p_A(2\pi i k) e_k \qquad k = 0, \pm 1, \pm 2, \ldots$$

The #'s $\lambda_k = p_A(2\pi i k)$ are eigenvalues of A.

The Adjoint of a Const. Coeff Diff. Op.

Look for $A^*: C^\infty_{per}(0,1) \to C^\infty_{per}(0,1)$, s.t.

$$\langle Au, v\rangle = \langle u, A^* v\rangle, \quad u, v \in C^\infty_{per}(0,1).$$

To calculate the adjoint, integrate-by-parts:

$$\left\langle a\frac{d^j u}{dx^j}, v\right\rangle = \int_0^1 a\frac{d^j u}{dx^j} v\, dx = \left[\frac{d^{j-1}u}{dx^{j-1}} v\right]_0^1 - \int_0^1 a\frac{d^{j-1}u}{dx^{j-1}}\frac{dv}{dx}\, dx$$

$$= (-1)^j a\int_0^1 u\frac{d^j v}{dx^j}\, dx = \left\langle u, (-1)^j \frac{d^j u}{dx^j} v\right\rangle$$

$$A^* = \left[\bar{a}_0 I - \bar{a}_1\frac{d}{dx} + \cdots + (-1)^m \bar{a}_m \frac{d^m}{dx^m}\right]$$

$$\rho_{A^*}(\lambda) = \bar{a}_0 - \bar{a}_1\lambda + \cdots + (-1)^m \bar{a}_m \lambda^m$$

$$[A^* e_k](x) = \rho_{A^*}(2\pi i k)\, e_k(x)$$

$$\rho_{A^*}(2\pi i k) = \left[\bar{a}_0 + \bar{a}_1\overline{(2\pi i k)} + \cdots + \bar{a}_m\overline{(2\pi i k)^m}\right]$$

$$= \overline{\rho_A(2\pi i k)}$$

$\Rightarrow$ The eigenvectors of A and A^* are the same (Fourier modes $e_k(x) = e^{2\pi i k x}$)

$\Rightarrow$ The eigenvalues of A are $\rho_A(2\pi i k)$ and eigenvalues of A^* are $\overline{\rho_A(2\pi i k)}$.

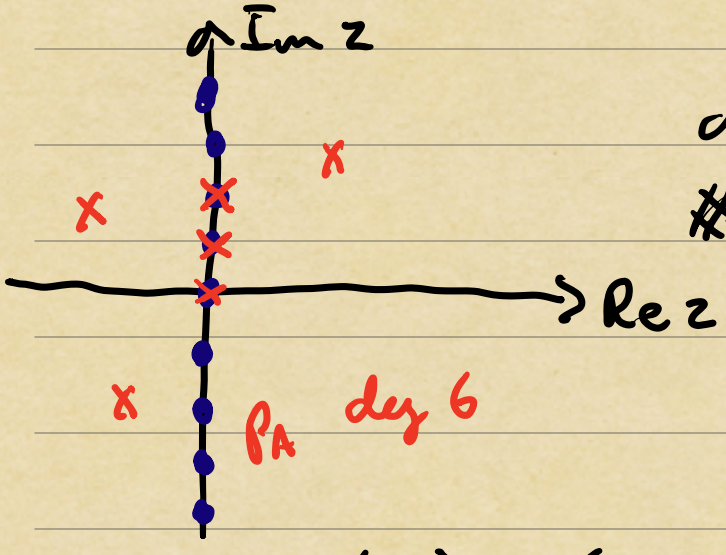
Homogeneous Eqn's

$$Au = 0, \qquad u \in C^\infty_{per}(0,1)$$

$$\Rightarrow N(A) = \{ e_k : \rho_A(2\pi i k) = 0 \}$$

$$\downarrow$$

$$Ae_k = \rho_A(2\pi i k) e_k = 0$$



$$\dim N(A) = \#\text{ roots of } \rho_A \cap \{2\pi i k\}_{k=-\infty}^{+\infty}$$

$$\Rightarrow N(A^*) = \{ e_k : \overline{\rho_A(2\pi i k)} = 0 \}$$

$$\downarrow$$

$$A^* e_k = \overline{\rho_A(2\pi i k)} e_k = 0$$

Inhomogeneous Equation

$$Au = f \qquad u \in C^\infty_{per}(0,1)$$

$$u(x) = \sum_{k=-\infty}^{+\infty} \hat{u}_k e_k(x), \qquad f(x) = \sum_{k=-\infty}^{+\infty} \hat{f}_k e_k(x)$$

match coeffs $\qquad \rho_A(2\pi i k)\,\hat{u}_k = \hat{f}_k \qquad k = 0, \pm 1, \pm 2, \dots$

Equivalent to solving a diagonal system of eqn's, thanks to working in the eigenvector basis for A.

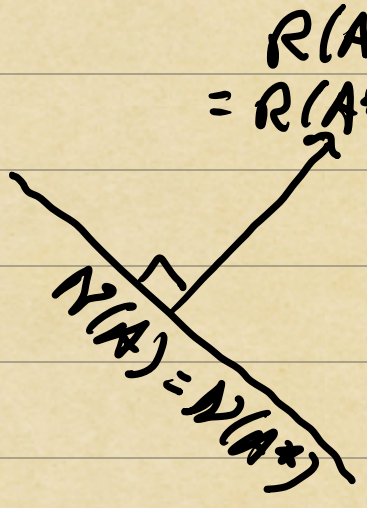
$$Ae_k = \rho_A(2\pi i k) e_k, \qquad A^* e_k = \overline{\rho_A(2\pi i k)}\, e_k$$

$$R(A) = \{Ae_k : \rho_A(2\pi i k) \neq 0\}$$

$$N(A^*) = \{e_k : \overline{\rho_A(2\pi i k)} = 0\}$$

$$R(A^*) = \{A^* e_k : \overline{\rho_A(2\pi i k)} \neq 0\}$$

$$N(A) = \{e_k : \rho_A(2\pi i k) = 0\}$$



General Solution ($k \in \mathbb{Z}$ iff $p_A(2\pi i k) = 0$)

If $f \perp N(A^*)$, then $\tilde{f}_k = 0$ for $k \in \mathbb{Z}$

$$u(x) = \sum_{k \in \mathbb{Z}} \hat{u}_k^{(\text{free})} e_k(x) + \sum_{k \in \mathbb{Z}^c} \frac{\hat{f}_k}{p_A(2\pi i k)} e_k(x)$$